

A PROJECT REPORT
on
“Emotion-Detection using CNN”

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BY

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DR. SRICHETA PARUI



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Abstract

This research paper examines the efficacy of Convolutional Neural Networks (CNNs) in recognizing facial emotions across seven distinct categories: anger, disgust, fear, happiness, neutral, sadness, and surprise. Using a dataset of 7,178 facial images, we implemented a CNN model that achieved an overall validation accuracy of 54.45%, with significant variation in classification performance across different emotion categories. Happiness and surprise were recognized with relatively high reliability (F1-scores of 0.74 and 0.70 respectively), while emotions such as fear and sadness presented greater classification challenges (F1-scores of 0.38 and 0.40). This paper presents a comprehensive analysis of the model's performance, identifies patterns in misclassification through confusion matrix analysis, and proposes potential enhancements to improve emotion recognition accuracy.

Keywords:

The main keywords for facial emotion recognition research would be:

1. Facial Emotion Recognition (FER)
2. Convolutional Neural Networks (CNN)
3. Computer Vision
4. Affective Computing
5. Deep Learning
6. Image Classification
7. Multi-class Classification
8. Emotion Detection
9. Machine Learning
10. Confusion Matrix Analysis

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Chapter 1

Introduction

Facial emotion recognition (FER) represents a crucial component of affective computing and human-computer interaction systems. The ability to automatically detect and classify human emotions from facial expressions has wide-ranging applications across various domains, including psychology, healthcare, marketing, security, education, and entertainment. This report analyzes a Convolutional Neural Network (CNN) model developed for recognizing seven basic facial emotions: anger, disgust, fear, happiness, neutral, sadness, and surprise.

Background and Significance

Human facial expressions communicate a wealth of information about emotional states, intentions, and reactions. While humans can intuitively recognize emotions displayed on faces, developing automated systems that can perform this task with comparable accuracy presents significant technical challenges. Recent advances in deep learning, particularly CNNs, have revolutionized computer vision tasks including facial emotion recognition.

Research Objectives

This project aims to develop and evaluate a CNN-based system for accurately classifying facial emotions across seven categories. The specific objectives include:

1. Designing an efficient CNN architecture suitable for facial emotion recognition
2. Training and optimizing the model using a diverse dataset of facial expressions
3. Evaluating the model's performance across different emotion categories



Chapter 2

Fundamental Concepts in Facial Emotion Recognition

1. Emotion Theories and Models

The field of facial emotion recognition is grounded in psychological theories of emotion. Paul Ekman's pioneering work established the concept of six universal basic emotions (anger, disgust, fear, happiness, sadness, and surprise) that are recognized across cultures, with neutral later added as a seventh category. This categorical approach to emotions forms the foundation of most facial emotion recognition systems. Alternative frameworks include dimensional models (valence-arousal space) and the Facial Action Coding System (FACS), which breaks down facial expressions into individual muscle movements called Action Units (AUs).

2. Image Preprocessing Techniques

Effective facial emotion recognition relies on standardized input data. Common preprocessing techniques include:

- **Face Detection:** Identifying and isolating facial regions within images using algorithms like Viola-Jones, HOG-based detectors, or deep learning approaches such as MTCNN or RetinaFace
- **Alignment:** Normalizing face orientation based on eye positions or facial landmarks
- **Resizing:** Standardizing images to fixed dimensions (e.g., 64×64 pixels in this study)
- **Normalization:** Scaling pixel values (typically to [0,1] range) to improve training stability
- **Data Augmentation:** Generating additional training samples through transformations like rotation, scaling, flipping, and brightness adjustments

3. Feature Extraction Methods

Feature extraction approaches in facial emotion recognition have evolved significantly:

- **Traditional Methods:** Early approaches relied on handcrafted features like Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), Gabor filters, and geometric features based on facial landmark distances
- **Deep Learning Features:** Modern approaches use CNNs to automatically learn hierarchical features from raw pixel data

4. Classification Algorithms

Various classification methods have been applied to facial emotion recognition:

- **Traditional Classifiers:** Support Vector Machines (SVM), Random Forests, and k-Nearest Neighbors were commonly used with handcrafted features
- **Deep Learning Approaches:** CNNs now dominate the field, offering end-to-end solutions that combine feature extraction and classification
- **Ensemble Methods:** Combining multiple classifiers to improve performance
- **Multi-stage Classifications:** Hierarchical approaches that first categorize broad emotion groups before fine-grained classification

5. Evaluation Metrics

Facial emotion recognition systems are evaluated using several metrics:

- **Accuracy:** Overall proportion of correctly classified images
- **Precision:** Ratio of true positives to all positive predictions for each emotion
- **Recall (Sensitivity):** Ratio of true positives to all actual instances of each emotion
- **F1-Score:** Harmonic mean of precision and recall
- **Confusion Matrix:** Visualization of classification performance showing patterns of correct classifications and misclassifications.

Literature Review

Evolution of Facial Emotion Recognition

Early Approaches (1990s-2000s)

The earliest automated facial emotion recognition systems emerged in the 1990s, utilizing geometric feature-based methods that tracked key facial points and measured their movements. Kobayashi and Hara (1997) developed one of the first systems using a neural network trained on displacement vectors of facial feature points. During the 2000s, appearance-based methods gained popularity, with Bartlett et al. (2003) using Gabor filters and Support Vector Machines to classify facial expressions.

Chapter 3

Model Architecture

The implemented CNN model follows a standard architecture for image classification tasks:

- Input layer accepting 64×64 pixel normalized facial images
- Multiple convolutional layers with increasing filter complexity
- Max pooling layers for dimensionality reduction
- Dropout layers to prevent overfitting
- Fully connected layers for final classification
- Output layer with 7 neurons (one for each emotion class)

The model was trained using the Adam optimizer with a sparse categorical cross-entropy loss function for 50 epochs, striking a balance between training time and model performance.

Dataset Overview and Distribution

The dataset consists of 7,178 facial images distributed across seven emotion classes:

The dataset exhibits significant class imbalance, with the "Disgust" category being severely underrepresented compared to other emotions. This imbalance likely contributed to the poor classification performance for this particular emotion class.

The dataset was divided into training (80%) and validation (20%) sets. All images were preprocessed by resizing to 64×64 pixels and normalizing pixel values to the [0,1] range to improve training

Training and Validation Process

The training process involved:

- Dataset splitting into 80% training and 20% validation sets
- Image preprocessing (resizing and normalization)
- Model training for 50 epochs using the Adam optimizer
- Use of sparse categorical cross-entropy as the loss function
- Regular evaluation on the validation set to monitor generalization

The significant gap between training accuracy (84.43%) and validation accuracy (54.45%) indicates potential overfitting, suggesting that the model learned patterns specific to the training data that did not generalize well to unseen examples.

Chapter 4

Overall Performance Metrics

The model achieved the following overall performance metrics:

The considerable difference between training and validation accuracy (approximately 30 percentage points) suggests that the model is overfitting to the training data. This indicates that while the model can memorize the training examples effectively, it struggles to generalize to new, unseen facial images.

Per-Class Performance Analysis

The model's performance varied significantly across the seven emotion classes:

Key observations:

- High-performing emotions: Happiness and Surprise were the most reliably detected emotions
- Moderate performance: Neutral expressions achieved middling performance
- Low-performing emotions: Fear, Sadness, and Anger were frequently misclassified
- Disgust: Despite relatively high precision, the recall was very low, likely due to the severe class imbalance (only 111 samples)

Confusion Matrix Analysis

The confusion matrix reveals several important patterns in misclassification:

High-Performing Emotions:

- Happy: Strong diagonal element (1,381 correct classifications), with some confusion with neutral (142)
- Surprise: Good performance (564 correct), with occasional confusion with fear (101)

Moderate-Performing Emotions:

- Neutral: Moderate performance (666 correct), but frequently confused with sad (197)

Low-Performing Emotions:

- Angry: Often misclassified as fear (140), neutral (139), and sad (153)
- Fear: Commonly confused with sad (195) and neutral (141)
- Sad: Frequently misidentified as neutral (271)
- Disgust: The least represented class, rarely predicted correctly (35 of 111 samples)

Key Confusion Patterns:

- Sad-Neutral Confusion: 271 sad images misclassified as neutral, 197 neutral images misclassified as sad
- Fear-Sad Confusion: 195 fear images misclassified as sad
- Angry-Sad Confusion: 153 angry images misclassified as sad

These patterns indicate that the model struggles to differentiate between emotions that share similar facial features, particularly between negative emotions like sadness, fear, and anger.

Chapter 5

Key Classification Challenges

Several factors contributed to the model's classification challenges:

1. **Class Imbalance:** The significant underrepresentation of the "Disgust" class (1.5% of the dataset) likely impaired the model's ability to learn this emotion effectively.
2. **Similar Facial Features:** Negative emotions (sad, fear, angry) share subtle facial expressions that are difficult to distinguish, even for humans in some cases.
3. **Neutral Confusion:** Many emotions in their milder forms can appear similar to a neutral expression, leading to frequent misclassifications.
4. **Limited Dataset Size:** Despite having 7,178 images, this is relatively small for deep learning standards, particularly when divided across seven classes.
5. **Model Complexity:** The chosen CNN architecture may not have sufficient capacity to capture the nuanced differences between similar emotions.
6. **Overfitting:** The significant gap between training and validation accuracy suggests the model memorized training examples rather than learning generalizable features.

Deployment Method

The model was deployed as a web application, allowing users to interact with the emotion recognition system through a user-friendly interface. The deployment architecture likely involves:

1. A backend server hosting the trained CNN model
2. A frontend web interface for user interaction
3. Image capture capabilities (via webcam or file upload)
4. Real-time or near-real-time emotion classification
5. Visual feedback displaying detected emotions

This deployment approach makes the emotion recognition technology accessible to users without requiring technical expertise or specialized hardware.

Visual Results and Interface

The web interface appears to include:

- Image input capabilities (webcam or file upload)
- Emotion classification results display
- Possibly confidence scores for each emotion class
- Visual indicators for the detected primary emotion
- User-friendly design for non-technical users

The interface provides immediate feedback on emotion detection, making it suitable for both research and practical applications.

Chapter 6

Conclusion

The facial emotion recognition CNN model demonstrates promising results for certain emotions (happiness, surprise) while facing challenges with others (fear, sadness, anger, disgust). The overall accuracy of 54.45% indicates room for significant improvement, particularly in distinguishing between similar negative emotions. By addressing class imbalance, enhancing the model architecture, implementing advanced training strategies, and expanding the dataset, future iterations of this system could achieve substantially better performance across all emotion categories.

The deployment as a web application makes this technology accessible and demonstrates its potential for real-world applications in fields such as human-computer interaction, mental health assessment, market research, and educational technology.

Recommendations for Improvement

Based on the analysis, several strategies could enhance the model's performance:

1. Address Class Imbalance:
 - Apply data augmentation techniques specifically to underrepresented classes
 - Utilize weighted loss functions to give more importance to minority classes
 - Consider oversampling techniques for the "Disgust" class
2. Enhance Model Architecture:
 - Implement transfer learning using pre-trained models like VGG, ResNet, or EfficientNet
 - Increase model depth or width to capture more nuanced features
 - Experiment with attention mechanisms to focus on discriminative facial regions
3. Regularization Techniques:
 - Increase dropout rates to combat overfitting
 - Apply batch normalization for more stable training
 - Implement early stopping based on validation performance
4. Feature Engineering:
 - Incorporate facial landmarks as additional input features
 - Consider multi-modal approaches combining image data with other signals
 - Extract action units (AUs) based on the Facial Action Coding System
5. Training Strategies:
 - Implement ensemble methods combining multiple models
 - Use curriculum learning to start with easy classifications before tackling more difficult ones
 - Explore contrastive learning to better differentiate similar emotions
6. Dataset Enhancements:
 - Expand the dataset, particularly for underrepresented emotions
 - Include more diverse subjects across different ages, ethnicities, and lighting conditions
 - Consider cross-dataset training to improve generalization

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Emotion-Detection Using CNN

INDIVIDUAL CONTRIBUTION REPORT

Emotion-Detection Using CNN

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Abstract:

This report presents a comprehensive analysis of a Convolutional Neural Network (CNN) model designed for facial emotion recognition across seven distinct emotional categories: anger, disgust, fear, happiness, neutral, sadness, and surprise. The model was developed and evaluated on a dataset of 7,178 facial images, with preprocessing including resizing to 64×64 pixels and normalization of pixel values. The CNN architecture incorporates multiple convolutional layers, max pooling, dropout for regularization, and fully connected layers for classification. Training was conducted using the Adam optimizer with sparse categorical cross-entropy loss over 50 epochs.

Individual contribution and findings:

Data preparation, preprocessing, model architect design, training, optimization, Performance Analysis, Web Application Development.

This facial emotion recognition project was conducted by Vanisha Nair (IT-4, Roll No.: 2206064) under the supervision of Dr. Sricheta Parui. The contributions encompassed several key aspects of the machine learning pipeline.

Individual contribution for project presentation and demonstration:

I independently prepared and presented the technical aspects, implementation details, and demonstration of the Emotion-Detection Using CNN. The presentation effectively communicated the project's objectives, methodology, and real-world applications, ensuring a clear and engaging demonstration.

Full Signature of Supervisor:

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Full signature of the student:

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